THE LIMIT OF A COMPATIBLE SYSTEM OF FINITE FLAT MODULES OVER A SURJECTIVE SYSTEM OF RINGS IS FINITE PROJECTIVE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The inverse limit pair of a system with surjective transition maps and locally nilpotent kernels is henselian. Lemma \ref{lemma-limit-henselian}, cited at proof line 3478. Allows the lifting result for finite projective modules over a henselian pair to be applied.
- Lifting of finite projective modules over a henselian pair. Lemma \ref{lemma-lift-finite-projective-module}, cited at proof lines 3479-3480. Produces the finite projective module over the limit ring mapping isomorphically to the first stage.
- Finite flat modules over the stages are finite projective. Lemma \ref{lemma-lift-projective}, cited at proof lines 3493-3494. Makes the lifting result applicable at every stage.
- Isomorphic finite projective modules lift along the transition maps. Lemma \ref{lemma-isomorphic-finite-projective-lifts}, cited at proof lines 3494-3495. Shows the comparison maps are isomorphisms at every stage.

Source reference. Original lemma 0D4B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16446–16446. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0CT7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16057–16057. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0D4A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16445–16445. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0D48: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16443–16443. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BC6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14269–14269. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A = \lim A_n$ be a limit of an inverse system (A_n) of rings. Suppose given A_n-modules M_n and A_{n+1}-module maps $M_{n+1} \to M_n$. Assume*

1. *the transition maps $A_{n+1} \to A_n$ are surjective with locally nilpotent kernels,*
2. *M_1 is a finite projective A_1-module,*
3. *M_n is a finite flat A_n-module, and*
4. *the maps induce isomorphisms $M_{n+1} \otimes_{A_{n+1}} A_n \to M_n$.*

Then $M = \lim M_n$ is a finite projective A-module and $M \otimes_A A_n \to M_n$ is an isomorphism for all n.

Proof. By Lemma 0CT7 the pair $(A, \mathrm{Ker}(A \to A_1))$ is henselian. By Lemma 0D4A we can choose a finite projective A-module P and an isomorphism $P \otimes_A A_1 \to M_1$. Since P is projective, we can successively lift the A-module map $P \to M_1$ to A-module maps $P \to M_2$, $P \to M_3$, and so on. Thus we obtain a map

$$P \longrightarrow M$$

Since P is finite projective, we can write $A^{\oplus m} = P \oplus Q$ for some $m \geq 0$ and A-module Q. Since $A = \lim A_n$ we conclude that $P = \lim P \otimes_A A_n$. Hence, in order to show that the displayed A-module map is an isomorphism, it suffices to show that the maps $P \otimes_A A_n \to M_n$ are isomorphisms. From Lemma 0D48 we see that M_n is a finite projective module. By Lemma 0BC6 the maps $P \otimes_A A_n \to M_n$ are isomorphisms. □